

DNS Guide for Thaghrah

Description

DNS (Domain Name System) is the "phonebook of the internet." It is a hierarchical and decentralized naming system that translates human-readable domain names (like example.com) into machine-readable IP addresses (like 192.0.2.1).

DNS makes the internet user-friendly — instead of remembering long IP addresses, we can use easy-to-remember names. It operates as a distributed database, with millions of DNS servers worldwide working together to resolve names quickly and reliably. DNS primarily uses **UDP port 53** for queries, but can fall back to **TCP port 53** for larger responses or zone transfers.

Key Components of DNS

1. DNS Resolution Process

When you type a domain name:

- Your device asks a **DNS Resolver** (usually provided by your ISP or public services like Google DNS).
- The resolver performs a series of queries through the DNS hierarchy:
 - **Root Servers** → **TLD Servers** (e.g., .com, .org) → **Authoritative Name Servers** for the specific domain.

2. Common DNS Record Types

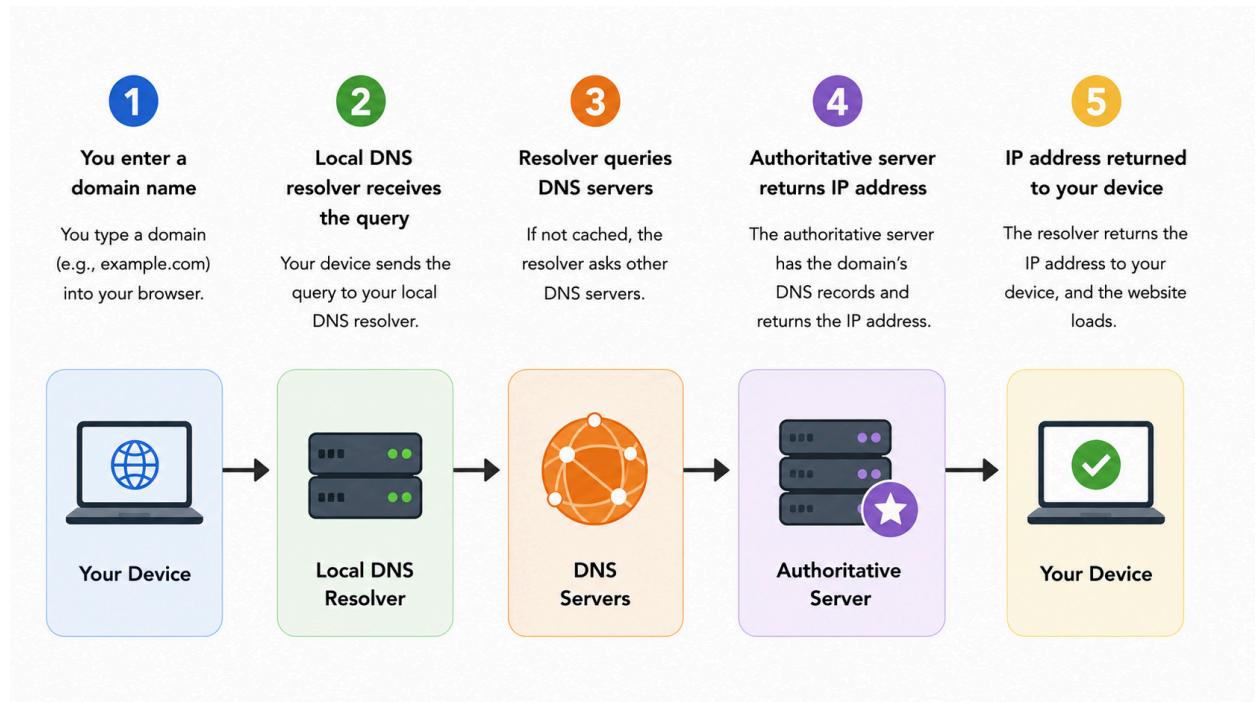
Type	Purpose	Example Value	Used For
 A	Maps a domain name to an IPv4 address.	<code>example.com. 3600 IN A 93.184.216.34</code>	Point a domain or subdomain to an IPv4 address.
 AAAA	Maps a domain name to an IPv6 address.	<code>example.com. 3600 IN AAAA 2001:db8::1</code>	Point a domain or subdomain to an IPv6 address.
 CNAME	Maps an alias domain name to another domain name.	<code>www.example.com. 3600 IN CNAME example.com.</code>	Create an alias (www → example.com).
 MX	Specifies the mail server(s) for a domain.	<code>example.com. 3600 IN MX 10 mail.example.com. example.com. 3600 IN MX 20 mail2.example.com.</code>	Receive email for the domain.
 TXT	Holds text information for a domain.	<code>example.com. 3600 IN TXT "v=spf1 include:_spf.example.com ~all"</code>	SPF, DKIM, domain verification, and other text-based info.
 NS	Specifies the authoritative name servers for a domain.	<code>example.com. 3600 IN NS ns1.example.com. example.com. 3600 IN NS ns2.example.com.</code>	Define which name servers are authoritative for the domain.
 PTR	Maps an IP address to a domain name (reverse DNS).	<code>34.216.184.93.in-addr.arpa. 3600 IN PTR example.com.</code>	Reverse DNS lookup (IP to domain name).
 SRV	Defines the location of services (host, port, priority).	<code>_sip._tcp.example.com. 3600 IN SRV 10 60 5060 sip.example.com.</code>	Locate services like VoIP, XMPP, SIP, etc.

3. Important Concepts

- **Recursive vs Iterative Queries**
- **Caching** — Results are stored temporarily to speed up future lookups
- **Authoritative vs Non-Authoritative Answers**
- **Zone Files** — Where DNS records are stored on servers

Visualizations

DNS Resolution Hierarchy / How DNS Works:



DNS Explained in 100 Seconds

<https://www.youtube.com/watch?v=UVR9IhUGAyU>